

Mathematical Physics I (Fall 2026): Homework #2 Solution

Due Oct. 2, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts]

1. Boas Chapter 3, Problem 5.37

(Note: For Problem 5.37, review the Examples in Boas Chapter 3, Section 5 in case you have not done it yet. Find the most relevant example and use the method depicted in there. Solve the problem in two different ways, following the two suggestions given. For the second method, start with the two points: $P(3, -2, 1)$ on the first line and $Q(5, -2, 0)$ on the second line.)

- Let $\mathbf{A} = (2, 1, -3)$ and $\mathbf{B} = (4, 1, -4)$ be the vectors along the two lines. Then, from Figure 5.8 and the accompanying equations, the distance between the two lines is $\left| \overrightarrow{PQ} \cdot \mathbf{n} \right| = \left| \overrightarrow{PQ} \cdot \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} \right| = \left| (2, 0, -1) \cdot \frac{1}{\sqrt{21}}(-1, -4, -2) \right| = 0$.

2. Boas Chapter 3, Problem 5.38

(Note: For Problem 5.38, find the acute angle between the lines.)

- With $\mathbf{A}$ and $\mathbf{B}$ defined in Problem 5.37, $\theta = \arccos \left(\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}| |\mathbf{B}|} \right) = \arccos \left(\frac{21}{\sqrt{14} \cdot \sqrt{33}} \right)$.

3. Boas Chapter 3, Problem 6.16

(Note: For Problem 6.16, find the inverse with two different methods — using Eq.(6.13) of Boas Chapter 3 and using the *Gauss-Jordan matrix inversion procedure*. Compare your results with the one found with a computer.)

- From Eq.(6.13),

$$M = \begin{pmatrix} -2 & 0 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 0 \end{pmatrix} \rightarrow \det(M) = 8 \text{ and } C = \begin{pmatrix} -2 & 6 & 4 \\ 1 & -3 & 2 \\ 1 & 5 & 2 \end{pmatrix} \rightarrow M^{-1} = \frac{1}{8} \begin{pmatrix} -2 & 1 & 1 \\ 6 & -3 & 5 \\ 4 & 2 & 2 \end{pmatrix}.$$

inverse {{-2,0,1},{1,-1,2},{3,1,0}}

 NATURAL LANGUAGE
 MATH INPUT

 EXTENDED KEYBOARD
 EXAMPLES
 UPLOAD
 RANDOM

Input

$$\begin{pmatrix} -2 & 0 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 0 \end{pmatrix}^{-1} \quad (\text{matrix inverse})$$

Result Decimal form ☐ Step-by-step solution ☒

$$\frac{1}{8} \begin{pmatrix} -2 & 1 & 1 \\ 6 & -3 & 5 \\ 4 & 2 & 2 \end{pmatrix}$$

4. Boas Chapter 3, Problem 7.27

- The matrix represents a rotation of $\frac{3\pi}{4}$ about $+z$ -axis with $\det(M) = 1$.

5. Boas Chapter 3, Problem 8.15

- Using the Wronskian in Eq.(8.5), you find

$$W = \begin{vmatrix} e^x & e^{ix} & \cosh x \\ e^x & ie^{ix} & \sinh x \\ e^x & -e^{ix} & \cosh x \end{vmatrix} = \begin{vmatrix} e^x & e^{ix} & \frac{e^x + e^{-x}}{2} \\ 0 & (i-1)e^{ix} & -e^{-x} \\ 0 & -2e^{ix} & 0 \end{vmatrix} = e^x \begin{vmatrix} (i-1)e^{ix} & -e^{-x} \\ -2e^{ix} & 0 \end{vmatrix} = -2e^{ix} \neq 0.$$

FROM THE MAKERS OF WOLFRAM LANGUAGE AND MATHEMATICA

wronskian(e^x, e^{ix}, coshx)

 NATURAL LANGUAGE
 MATH INPUT


 EXTENDED KEYBOARD
 EXAMPLES
 UPLOAD


Assuming i is the imaginary unit | Use i as a variable instead

Input interpretation:

	e^x	
Wronskian	e^{ix}	with respect to x
	$\cosh(x)$	

$\cosh(x)$ is the hyperbolic cosine function
 i is the imaginary unit

Result:

$$-2e^{ix}$$

6. Boas Chapter 3, Problem 9.17

- (c) $A = A^\dagger$ and $B = B^\dagger \rightarrow$ from a complex counterpart of Eq.(9.11), $(AB)^\dagger = B^\dagger A^\dagger = BA = AB$ holds (i.e., AB is Hermitian) if and only if $[A, B] = 0$.
- (d) $A^{-1} = A^\dagger$ and $B^{-1} = B^\dagger \rightarrow$ from Eq.(9.12), and from a complex counterpart of Eq.(9.11), $(AB)^{-1} = B^{-1}A^{-1} = B^\dagger A^\dagger = (AB)^\dagger$ holds (i.e., AB is unitary).

7. Boas Chapter 3, Problem 11.44

• From the characteristic equation, two eigenvalues $\lambda_1 = -7$ and $\lambda_2 = 3$ appear that correspond to eigenvectors $\mathbf{r}_1 = \frac{1}{5\sqrt{2}} \begin{pmatrix} -3-4i \\ 5 \end{pmatrix}$ and $\mathbf{r}_2 = \frac{1}{5\sqrt{2}} \begin{pmatrix} 5 \\ 3-4i \end{pmatrix}$, respectively. Then it is straightforward to show that

$$U^{-1}HU = -\frac{1}{50} \begin{pmatrix} 3-4i & -5 \\ -5 & -3-4i \end{pmatrix} \begin{pmatrix} -2 & 3+4i \\ 3-4i & -2 \end{pmatrix} \begin{pmatrix} -3-4i & 5 \\ 5 & 3-4i \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

yields the diagonal matrix of eigenvalues.



$$(-1/50) * \{\{3-4i, -5\}, \{-5, -3-4i\}\} * \{\{-2, 3+4i\}, \{3-4i, -2\}\} * \{\{-3-4i, 5\}, \{5, 3-4i\}\}$$

NATURAL LANGUAGE
 MATH INPUT
 EXTENDED KEYBOARD
 EXAMPLES
 UPLOAD
 RANDOM

Assuming "/" is referring to math | Use "-1/50" as referring to math instead

Assuming i is the imaginary unit | Use i as a variable instead

Input

$$\left(-\frac{1}{50} \begin{pmatrix} 3-4i & -5 \\ -5 & -3-4i \end{pmatrix}\right) \begin{pmatrix} -2 & 3+4i \\ 3-4i & -2 \end{pmatrix} \begin{pmatrix} -3-4i & 5 \\ 5 & 3-4i \end{pmatrix}$$

i is the imaginary unit

Result

$$\begin{pmatrix} -7 & 0 \\ 0 & 3 \end{pmatrix}$$

8. Boas Chapter 3, Problem 12.16

• Following the same method as in Eqs.(12.10)-(12.14) or Eqs.(12.18)-(12.20), we can write the equations of motion in matrix form

$$\lambda \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 & -2 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

where $\lambda = \frac{m\omega^2}{k}$. The equation yields the eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 7$ and the characteristic frequencies $\omega_1 = \sqrt{\frac{2k}{m}}$ and $\omega_2 = \sqrt{\frac{7k}{m}}$, corresponding to eigenvectors $\mathbf{r}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ of symmetric mode and $\mathbf{r}_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ of antisymmetric mode, respectively.

9. The polarization of electromagnetic waves and the properties of optical filters can be described using vectors and matrices.

(a) A beam of linearly polarized light is described by the two-component vector

$$\mathbf{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix}, \quad (1)$$

where E_x and E_y are the amplitudes of the electric field along two perpendicular directions, x and y . Meanwhile, an ideal linear polarizer whose transmission axis makes an angle θ with the x -axis is represented by the matrix

$$P(\theta) = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix}.$$

Show that $P(\theta)^2 = P(\theta)$, and explain why this property is expected for an *ideal* polarizer.

(b) Find the eigenvalues and normalized eigenvectors of $P(\theta)$, and interpret the two eigenvectors physically.

(c) Consider two ideal polarizers. The first has its transmission axis along the x -axis, while the second is rotated by an angle θ relative to the first. Then the electric field after passing through the two polarizers can be written as $\mathbf{E}_{\text{out}} = P(\theta)P(0)\mathbf{E}_{\text{in}}$. Given the incident light in Eq.(1), find $\mathbf{E}_{\text{out}}$ explicitly. Show that the intensity of the transmitted light is $I_{\text{out}} = I_0 \cos^2 \theta$, where I_0 is the intensity after the first polarizer.

(d) Now consider three ideal polarizers. The first and last polarizers are crossed, while the middle polarizer is oriented at an angle θ relative to the x -axis:

$$P(0), \quad P(\theta), \quad P\left(\frac{\pi}{2}\right).$$

Show that although the first and last polarizers are crossed and would completely block the light if placed directly together, a nonzero intensity can emerge when the intermediate polarizer is inserted. Find the transmitted intensity as a function of θ and determine the angle at which the transmitted intensity is maximized.

(Note: For (c) and (d), you may want to briefly review the freshman physics textbooks such as Halliday & Resnick Chapter 33.4.)

- (b) The first eigenvector corresponding to an eigenvalue 1 is the polarization direction transmitted by the polarizer. The second eigenvector corresponding to an eigenvalue 0 is perpendicular to the transmission axis, completely blocked by the polarizer.
- (c) The electric field of the transmitted light is

$$\mathbf{E}_{\text{out}} = P(\theta)P(0)\mathbf{E}_{\text{in}} = E_x \begin{pmatrix} \cos^2 \theta \\ \cos \theta \sin \theta \end{pmatrix},$$

which straightforwardly leads to $I_{\text{out}} = I_0 \cos^2 \theta$.

- (d) With an additional polarizer, the transmitted light is

$$\mathbf{E}_{\text{out}} = P\left(\frac{\pi}{2}\right)P(\theta)P(0)\mathbf{E}_{\text{in}} = E_x \begin{pmatrix} 0 \\ \cos \theta \sin \theta \end{pmatrix},$$

which yields the intensity $I_{\text{out}} = \frac{1}{4}I_0 \sin^2(2\theta)$.

10. In the class we discussed the Gram-Schmidt orthonormalization process for a linear vector space and for a general vector space.

(a) In Example 6 of Boas Chapter 3, Section 14 — which we briefly discussed in class but left for you to work through — with an inner product defined as

$$\langle f|g \rangle = \int_{-1}^1 f^*(x)g(x)dx,$$

one can start with the functions $f_i = x^i$ ($i = 0, 1, 2, 3$) and construct a set of orthonormal polynomials P_i that satisfy the orthonormality condition on the interval $-1 \leq x \leq 1$,

$$\int_{-1}^1 P_m(x)P_n(x)dx = \delta_{mn}$$

(see also Eq.(8.4) of Chapter 12, but note a different normalization factor). We later identify this set of functions as the *Legendre polynomials*. Starting from Eq.(14.10), follow the procedure step by step and find for yourself the first four members of P_i .

(b) Consider a different set of functions defined with a similar inner product as in (a), but over a shifted interval $0 \leq x \leq 1$. The orthonormality condition is now given by

$$\int_0^1 P'_m(x)P'_n(x)dx = \frac{1}{2m+1} \delta_{mn}.$$

Find the first three members in the set of orthonormal polynomials P'_i (the sign ' does not mean differentiation). We refer to this set of functions as the *shifted Legendre polynomials*.

(c) Now, with a new inner product defined as

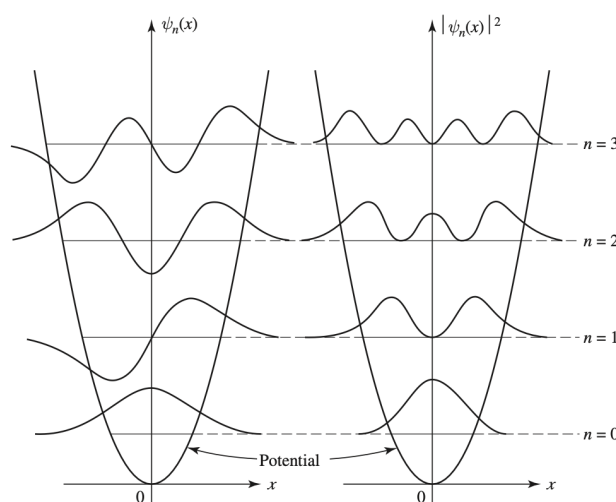
$$\langle f|g\rangle = \int_{-\infty}^{\infty} f^*(x)g(x)e^{-x^2} dx, \quad \text{where } e^{-x^2} \text{ is the "weighting" function,}$$

find the first three members in the set of orthonormal polynomials H_i that satisfy the orthonormality condition on the interval $-\infty \leq x \leq \infty$,

$$\int_{-\infty}^{\infty} H_m(x)H_n(x)e^{-x^2} dx = \delta_{mn}\sqrt{\pi} 2^m m!$$

(see also Eq.(22.15) of Chapter 12). We refer to this set of functions as the *Hermite polynomials*.

(d) Discuss briefly where the (associated) Legendre polynomials and the Hermite polynomials appear in quantum mechanics or in other physics research.



[Adapted from Griffiths & Schroeter, Chapter 2.3]

(Note: For (c), you may simply use the value of the so-called Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ without proof, although it can be easily found using the techniques in Boas Chapter 5, Section

4, or in Problem 9.4 of Chapter 11. The Gaussian integral appears frequently also in elementary calculus textbooks such as Stewart (Chapter 15.3) or 김홍중 (Chapter 14.4). However, all other integrals such as $\int_{-\infty}^{\infty} x^2 e^{-x^2} dx$ must be evaluated explicitly, for example, by using integration by parts or the technique in Problem 12.16 of Boas Chapter 4. For (d), you may want to briefly review the textbooks in quantum mechanics such as Griffiths & Schroeter. Once again, you are simply asked to come up with 2-3 sentences about how these polynomials are used, not the detailed physical or mathematical discussions. *Astronomy majors* are highly encouraged to look for the use cases of Legendre polynomials in astrophysics and cosmology.)

- (b-1) Following the notation in Example 6 of Boas Chapter 3, Section 14, but with $P'_i = e_i$, first we find $P'_0 = p_0 = f_0 = 1$ as $\|p_0\|^2 = \int_0^1 dx = 1$.
- (b-2) Then $p_1 = f_1 - P'_0 \int_0^1 P'_0 f_1 dx = x - 1 \cdot \int_0^1 x dx = x - \frac{1}{2} \rightarrow$ therefore $P'_1 = 2x - 1$, because $\|p_1\|^2 = \int_0^1 (x - \frac{1}{2})^2 dx = \frac{1}{12}$ while $\|P'_1\|^2$ should be $\frac{1}{2 \cdot 1 + 1} = \frac{1}{3}$.
- (c-1) Again, following the notation in Example 6 of Boas Chapter 3, Section 14, but now with $H_i = e_i$, we find $H_0 = p_0 = f_0 = 1$ as $\|p_0\|^2 = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} = \delta_{00} \sqrt{\pi} \cdot 2^0 \cdot 0!$.
- (c-2) Then $p_1 = f_1 - H_0 \int_{-\infty}^{\infty} H_0 f_1 e^{-x^2} dx = x - 1 \cdot \int_{-\infty}^{\infty} x e^{-x^2} dx = x \rightarrow$ therefore $H_1 = 2x$, because $\|p_1\|^2 = \int_{-\infty}^{\infty} x^2 e^{-x^2} dx = \left[x \left(-\frac{1}{2} e^{-x^2} \right) \right]_{-\infty}^{\infty} + \frac{1}{2} \int_{-\infty}^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ while $\|H_1\|^2$ should be $\delta_{11} \sqrt{\pi} \cdot 2^1 \cdot 1! = 2\sqrt{\pi}$.
- (c-3) The list of the first few normalized Hermite polynomials are given in Eq.(22.13) of Chapter 12, or in p.804 (the answer to Problem 22.4 of Chapter 12).